

The Ideal Data Ecosystem for Human Capital Valuation at CACEIS

Part 1 - Designing from a blank sheet: data sources, architecture, analytics, and governance for a finance-grade human capital valuation system.

The question CACEIS now faces is not whether human capital should be valued, but how — with what data, what models, and what level of auditability. Between ISO 30414:2025 and the ESRS S1 standard under the CSRD, the regulatory bar has risen to match the rigor of financial reporting.

Asset servicing is a business whose physical balance sheet is small but whose delivered value rests almost entirely on the expertise, judgment, and continuity of its people.

In this first part, we describe the data ecosystem we would design from scratch to measure, monitor, and steer that value. Part 2 will then confront this ideal with the data CACEIS has actually provided.

1. Conceptual Framework for Human Capital Valuation

Traditional accounting treats salaries, training, and benefits as expenses that depress operating income.

We propose the opposite: treat the workforce as a productive asset whose value can be estimated, tracked over time, and tied to the creation of economic profit. Three complementary layers make this shift operational.

- The first layer is productive value: what each full-time equivalent adds to revenue once external costs are removed. Human Capital Value Added (HCVA) sits squarely here and gives a CFO a defensible per-head productivity benchmark.
- The second layer is economic-profit value: how talent moves Economic Value Added (EVA) after the cost of capital. Recent studies referenced in our background research show that HR drivers, engagement and training investment in particular, explain more than half the variance in EVA for service firms, with beta coefficients of 0.42 for engagement and 0.36 for training.
- The third layer is strategic-alpha value: the degree to which CACEIS' workforce lets it outperform peers and absorb market shocks. This is the logic JP Morgan Asset Management has productized through its Human Capital Factor: companies with stable, well-trained teams generate measurable alpha versus standard benchmarks. 

On top of these three layers we recommend a balance-sheet view using the Lev-Schwartz model, refined with the Flamholtz probabilities of promotion and attrition.

The Human Asset Value it produces (the discounted present value of expected future earnings across the workforce) is imperfect, as it rewards tenure more than contribution. But it gives leadership a single figure to track, and, crucially, it makes M&A talent due diligence (the kind BNP Paribas publicly applied to the AXA IM acquisition in 2025) a quantitative exercise rather than a narrative one.

Value, in our framework, has four dimensions:

- Financial (contribution to revenue and economic profit)
- Capability (skills inventory and depth on critical roles)
- Relational (network health, knowledge transfer, client continuity)
- Sustainability (wellbeing, inclusion, retention of talent).

No single indicator captures all four, which is why the ecosystem that follows is deliberately wider than any HRIS alone can supply.

2. Ideal Data Sources

2.1 Internal data

The backbone of the system is the HRIS record: demographics, contract type, FTE count by country and business line, compensation, start and end dates. This is non-negotiable and must be clean enough on day one to meet ISO 30414:2025 composition requirements. Around it, we would deploy eight complementary internal streams.

Performance and objectives data from the OKR or performance management system connects individual contribution to business outcomes.

- Recruitment funnel data from the ATS: time-to-fill, source quality, offer acceptance rate, 90-day retention (lets us measure the cost and effectiveness of talent acquisition, not just its activity).
- Learning and development data captured in the LMS, ideally tagged against a formal skills taxonomy, reveals where capability is being built and where it is decaying. Internal mobility data (promotions, lateral moves, rotations) surfaces the organization's talent liquidity: a leading indicator of both engagement and succession depth.

Engagement and sentiment data is where we would push hardest beyond the annual survey.

- Quarterly pulse surveys combined with event-triggered micro-surveys after key milestones (onboarding, project closure, manager change) give a signal that moves fast enough to act on.
- Collaboration metadata from Microsoft 365 and Teams, feeding an Organizational Network Analysis layer, maps who collaborates with whom, how densely, and where the informal brokers sit, without ever reading message content.
- Absence and aggregated wellbeing data warn us of burnout before it shows up as resignation.
- Finally, operational performance data: error rates, SLA compliance, ticket resolution times, assets serviced per FTE: closes the loop between human capital investment and the client-facing quality on which CACEIS is ultimately judged.

2.2 External data

No internal dataset can tell us whether a 12% voluntary turnover rate is good or bad for a custodian operating across France and Luxembourg. External benchmarks anchor every metric we produce. We recommend four families of external data.

- Compensation benchmarks from Mercer, Willis Towers Watson, and the McLagan survey (financial-services specialist) tell us whether pay is competitive by job family and seniority.
- Labor market data from LinkedIn Talent Insights, Glassdoor, Lightcast, and Revelio Labs show us how quickly competitors are hiring, in which skills, and where salary pressure is building.
- A unified skills taxonomy, ideally ESCO mapped to SFIA for technology roles and to the internal CACEIS job catalog, makes skills inventories comparable across countries and against the market.
- And peer disclosures from the 2025 CSRD filings of BNP Paribas, Société Générale, State Street, BNY, Northern Trust, and Crédit Agricole S.A. itself give us a structured peer benchmark on every ESRS S1 indicator.

2.3 Complementary data beyond the current scope

Four further sources, less obvious but high-leverage, would round out the ecosystem.

- Client-to-employee linkage data. Matching client retention and client satisfaction to the stability of the teams that serve them is, in our view, the single most powerful analytical move available to an asset servicer.
 - It makes the business case for human capital investment concrete: if a ten-point drop in turnover in fund accounting correlates with a three-point rise in client retention, the training budget writes itself.
- AI-collaboration telemetry. Microsoft's 2025 annual report introduced the "human-agent ratio" (the share of employee time augmented by AI copilots) as a core productivity disclosure.
 - We would track Copilot adoption, prompt volume, and automation rate per job family to measure how quickly CACEIS teams are moving up the value chain.
- Knowledge capital signals. Contributions to internal knowledge bases, documented procedures, published methodologies, and cross-training sessions all represent intellectual capital that outlives individual employees.
 - A simple contribution index per team is a leading indicator of resilience to departures.
- Fourth, alumni data. Former employees are a predictive dataset in their own right: where they went, how long they stayed, what their next employer paid them.
 - It reveals both the real market value of CACEIS-trained talent and the competitive pressure building in specific functions.

The table below summarizes the full source map.

Source category	Representative feeds	Primary use
Core HRIS	Workday or SAP SuccessFactors: demographics, FTE, contract type, compensation, tenure, organizational hierarchy.	Workforce composition; HCVA denominator; regulatory disclosures.
Talent lifecycle	ATS data (time-to-fill, source quality, offer acceptance, 90-day retention); mobility events; performance ratings; 360 feedback.	Cost-to-hire, internal mobility, succession depth.
Capability	LMS records; certifications; skill tags mapped to a unified taxonomy (ESCO + SFIA); AI-copilot adoption telemetry.	Skills-gap index; training ROI; human-agent ratio.
Engagement & sentiment	Quarterly pulse surveys; event-triggered micro-surveys (post-onboarding, post-project); anonymized eNPS; internal chat sentiment (aggregated only).	Early warning on disengagement; attrition-risk inputs.
Collaboration (ONA)	Microsoft 365 metadata via Viva Insights or Worklytics: meeting load, collaboration hours, network density, brokerage scores. Metadata only, never content.	Network health; burnout risk; knowledge-broker identification.
Operational & client	Assets serviced per FTE; SLA compliance; error and incident rates; client NPS; client retention by servicing team.	Links human capital to client outcomes — the core CACEIS use case.
External benchmarks	Mercer, Willis Towers Watson, McLagan compensation surveys; LinkedIn Talent Insights; Lightcast job postings; 2025 ESRS S1 peer filings (BNP Paribas, State Street, BNY, Northern Trust, Crédit Agricole SA).	Market calibration; HCF-style relative scoring.

3. Collection Architecture, Tools & Governance

We recommend a hub-and-spoke architecture with a dedicated Human Capital Data Lakehouse at the center.

The HRIS (Workday or SAP SuccessFactors) and the ATS are the authoritative systems of record. They feed the lakehouse through change-data-capture pipelines so that every update (a promotion, a resignation, a new hire) propagates in near real time rather than in monthly batches.

The LMS, performance tools, collaboration platforms (Microsoft 365, Teams, Viva Insights), engagement tools (Glint, Peakon, or Culture Amp), and the talent-intelligence layer (Eightfold or Gloat) each stream into the same lakehouse through documented connectors. External data

(compensation surveys, peer filings, labor market feeds) arrives via scheduled API pulls or vendor uploads into clearly marked external zones.

On top of the lakehouse sits a transformation layer (dbt or equivalent) that produces the canonical tables:

- One row per employee-month, one row per job-family-month, one row per business-unit-month.
- A semantic layer (Cube or the dbt Semantic Layer) enforces a single definition of every metric. "Turnover" will mean the same thing whether it surfaces in the CFO's EVA deck or in the CHRO's board pack (ending the most common failure mode of human capital reporting programs).
- Dashboards are served through Power BI for operational users and Tableau or ThoughtSpot for analytical power users.

Governance runs through a Human Capital Data Steering Committee chaired jointly by the CHRO and the CFO, with permanent membership from the CDO, the DPO, the Head of Internal Audit, and the business-unit heads.

Every critical attribute (salary, performance rating, engagement score, termination reason) has a named business owner and a technical steward, following the BCBS 239 principles already familiar to CACEIS' risk function.

A central data dictionary, published in Collibra or Atlan, is the single reference for definitions, lineage, and quality rules.

4. KPIs & Analytical Capabilities

We propose a three-tier KPI framework. Each tier serves a different audience and is built from the tier below, so that the regulatory disclosures, the financial metrics, and the predictive analytics all rest on the same governed data.

- Tier 1 covers regulatory disclosures: the ISO 30414:2025 requirements and ESRS S1 mandatory data points.
- Tier 2 covers financial valuation, translating human capital into the CFO's language through HCROI, HCVA, HAV, and the isolated workforce contribution to EVA.
 - A dedicated line "workforce contribution to economic profit" should appear in internal P&L reviews alongside traditional margin analysis.
- Tier 3 is where real alpha is generated: predictive attrition scores by team and role, skills-gap indices against a two-year strategic roadmap, ONA collaboration-health indicators, an innovation index built from cross-functional connectivity, and a CACEIS-specific Human Capital Factor score calibrated on peer benchmarks.

Tier	Indicators	Primary audience & purpose
Tier 1 Regulatory	Workforce composition (country, age, gender, contract); total HC cost; unadjusted gender pay gap; voluntary and total turnover; training hours and investment per FTE; workplace incident rate; collective bargaining coverage.	CSRD/ESRS S1 and ISO 30414:2025 disclosures. Audit-ready. Feeds the sustainability report and the board's ESG committee.
Tier 2 Financial	HCROI; HCVA per FTE; Human Asset Value (Lev-Schwartz with Flamholtz adjustments); isolated workforce contribution to EVA; cost-to-hire and cost-to-replace by job family.	CFO, ExCom, investor relations. Translates workforce into economic profit. Informs budget arbitrage and M&A talent due diligence.
Tier 3 Strategic & predictive	Predictive attrition score by team and role; skills-gap index against 2-year roadmap; ONA collaboration-health score; innovation index; client-retention-to-employee-retention correlation; CACEIS Human Capital Factor (HCF) score.	CHRO, business-unit heads, strategic workforce planning. Answers what is going to happen, not what has happened. Drives intervention before value leaks.

Three analytical capabilities emerge from this ecosystem that CACEIS cannot produce today.

- The first is predictive attrition at team level, with lead times of three to six months (long enough to intervene).
- The second is a client-retention-to-employee-retention correlation per servicing line, turning the argument for human capital investment into a quantified business case rather than an HR claim.
- The third is a CACEIS HCF score comparable across business units and against peers, which can feed both internal capital allocation and external investor communication.

5. Continuous Data Collection Strategy

Annual surveys are a diagnostic from the past. The ecosystem we describe is built to run on a continuous rhythm with three clocks in parallel.

- The real-time clock covers collaboration metadata, operational performance, and lifecycle events: a hire, a departure, a promotion, a role change.
 - These flow through event streams and update the lakehouse within minutes.

- The weekly-to-monthly clock covers pulse surveys, engagement micro-surveys, dashboard KPI refreshes, and attrition-model rescoring. It gives HR business partners and managers a cadence short enough to act on.
- The annual clock retains the deep diagnostic instruments: full engagement survey, 360-degree reviews, strategic workforce planning, salary review (that require reflection rather than reaction).

What makes the system genuinely continuous is not the speed of any one feed, but the consistency of the semantic layer across all of them: the same employee ID, the same skill taxonomy, the same definition of turnover, whether the data arrives every five seconds or once a year.

Each metric also carries a timestamp and a version tag, so longitudinal analysis remains possible even when methodologies evolve.

Automated data-quality checks run at ingestion: completeness, referential integrity, anomaly detection and route exceptions to the relevant steward. This is how a data ecosystem built for real-time analytics also becomes audit-ready for CSRD limited and reasonable assurance.

6. Technological Infrastructure

The infrastructure rests on five layers, each built on the assumption that the system will grow and be audited.

- The ingestion layer uses change-data-capture pipelines from transactional systems (Workday, SuccessFactors, ATS), event streaming for collaboration and operational data, and scheduled pulls for external feeds.
 - Where possible, we standardize on managed connectors (Fivetran or Airbyte) to keep the pipeline estate small.
- The storage and transformation layer is a lakehouse on Snowflake, Databricks, or Microsoft Fabric. Given the prevalence of the Microsoft 365 stack in CACEIS' collaboration environment (and the ONA logic that depends on it)
 - Fabric has a natural integration advantage. Transformations and tests run in dbt, with a medallion architecture (bronze, silver, gold) that makes lineage and data quality inspectable at every step.
- The intelligence layer is an MLOps platform (Databricks ML, Azure ML, or SageMaker) hosting the attrition-risk model, the skills-matching model, the client-retention-linkage model, and the sentiment-analysis pipelines.
 - Every production model has a model card, a drift monitor, and a scheduled retraining cycle: requirements that are no longer optional under the EU AI Act.
- The semantic and governance layer combines a data catalog (Collibra or Atlan), a shared metric store (Cube or dbt Semantic Layer), and an access-control matrix wired through the corporate identity provider with role-based and attribute-based rules.

- The consumption layer serves Power BI for executive dashboards, Tableau or ThoughtSpot for self-service analytics, and an embedded-analytics API for internal products that surface HR metrics to managers.
- Security cuts across every layer: encryption at rest and in transit as a default, tokenization of directly identifying fields (national ID, bank account) in analytical environments, and a zero-trust access model where no query runs outside a controlled workspace.
 - Access is logged, reviewed quarterly, and tied to an automated joiner-mover-leaver process.

7. GDPR Compliance & Ethical Governance

A system that aggregates collaboration metadata, sentiment scores, and predictive attrition models is, under European law, one of the most sensitive data estates a bank can operate. We treat compliance and ethics as design constraints, not as a final review stage.

The legal basis for processing is the execution of the employment contract together with CACEIS' legitimate interest in workforce management.

- We do not rely on employee consent, which is rarely valid in an employer-employee relationship under the EDPB's guidance.
- Every processing activity is recorded in the Article 30 register and supported by a Data Protection Impact Assessment where the risk profile requires one, which, for ONA and predictive attrition, it always does.

Privacy-by-design is the default at every analytical layer.

- ONA and sentiment analytics apply a minimum group size of five, so that no individual score is ever visible.
- Direct identifiers are tokenized in analytical zones, and re-identification requires a second, logged access through a break-glass procedure.
- Sensitive categories of data (health, union membership, biometric information) are either excluded or processed only by named teams under a separate DPIA.

The EU AI Act classifies People Analytics systems used in recruitment, performance evaluation, or promotion decisions as high-risk.

- Every model that informs such a decision carries a model card documenting its training data, input features, measured bias, and known limitations; a human-in-the-loop for every individual decision; and an annual algorithmic audit.
- We recommend an Ethics Committee: the DPO, a CHRO representative, a business ethics officer, and an independent external member, that reviews every high-risk model before deployment and annually thereafter.

Employee transparency is not optional. CACEIS employees should be able to see, in plain language, which data is collected about them, which models use it, and how to contest an automated decision.

- We recommend a dedicated section of the employee portal, and a works council consultation before any new data source goes live
 - Because the “Code du travail” requires it in France, and because it is the right thing to do.
 - Retention rules are codified in the catalog and enforced automatically: most HR data is kept for five years after departure, specific categories (training certificates, disciplinary records) for longer per local law, and everything that falls outside those windows is purged on a monthly schedule with a full audit log.

Closing note

The ecosystem described in this document is ambitious but internally consistent: every data source has a business purpose, every metric feeds a decision, every decision is traceable to a governed definition, and every algorithm is auditable.

It is designed to survive both a CSRD auditor and an EU AI Act inspection and, more importantly, to give CACEIS' leadership a live reading of the asset on which its entire franchise rests.

Part 2 of this deliverable will hold this ideal against the data CACEIS has actually provided and propose a pragmatic path from today's reality toward this target.